ETS2-ATS Instrument Cluster User guide

A stand-alone digital instrument cluster for Euro Truck Simulator 2 and American Truck Simulator, shown in its own window that you can place anywhere.

Version 1.5.0 for Windows 10 and 11, and for Android tablets and phones



Screenshots in this guide were taken with demonstration values.

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1. What it is

The ETS2-ATS Instrument Cluster is a digital dashboard for Euro Truck Simulator 2 and American Truck Simulator in the style of a modern truck cluster: a speedometer with a needle, gear, rev strip, fuel and AdBlue gauges, consumption, range, cruise-control setting, arrival time, clocks, odometer and trip, plus warning lamps.

It is a separate program from the ETS2-ATS GPS apps, so you can put the cluster on a second monitor, below the game, or beside the GPS window, wherever you like. It only **reads** information from the game (through the TruckSim GPS Telemetry Server) and never sends anything to the game.

There is also an **Android version** with the same layout, ideal for an old tablet mounted beside your monitor (see section 9).

By default the cluster uses the game's own units: in **American Truck Simulator mph and miles**, consumption in **miles per gallon** and temperatures in **°F**; in ETS2 km/h, kilometres, l/100km and °C. The **Units** menu can set **Metric** or **Imperial** instead, in either game. In ATS it shows **DEF** (diesel exhaust fluid) in place of AdBlue.

With the **ETS2-ATS GPS** Windows app (1.12.0 or later) and its voice control on, you can say **"GPS, minimize"** and **"GPS, restore"** to minimize or bring back the GPS window and the cluster together, or **"GPS, minimize the cluster"** / **"GPS, restore the cluster"** for the cluster alone (use your own wake word). Restoring does not take the keyboard from the game.

What you need

- Windows 10 or 11 (64-bit). Nothing else to install: the program carries everything it needs.
- Euro Truck Simulator 2 or American Truck Simulator with the **TruckSim GPS Telemetry Server** installed and running. This is the same source the ETS2-ATS GPS apps use. The cluster asks it for the truck's data about ten times a second.
- The game should be run **windowed or borderless** if you want the cluster window to stay on top of it on the same monitor. On a second monitor any mode works.

> **TIP** The cluster does not need the ETS2-ATS GPS app. They are independent: use either one, or both side by side.

2. Installing

2.1 With the setup program (recommended)

1. Download **Ets2ClusterSetup-1.5.0.exe** from the **Releases** page of the ETS2-ATS-Instrument-Cluster repository on GitHub.
2. Double-click the file. It installs for your user only, so **no administrator rights are needed**.
3. Choose whether to add a desktop shortcut, and whether the cluster should start automatically when you log in to Windows. Both are optional and off by default.
4. Tick *Start the instrument cluster now* on the last page, or start it later from the Start Menu: **ETS2-ATS Instrument Cluster**.

The setup program also puts a shortcut to this guide in the Start Menu folder. To remove the program, use *Settings* > *Apps* > *Installed apps*; the uninstaller asks whether to delete your saved window position and settings too.

2.2 Without installing (portable)

Download **Ets2Cluster-portable-1.5.0.zip**, unpack it to any folder (for example on a USB stick) and run **Ets2Cluster.exe**. The settings are still kept in `%APPDATA%\ETS2-Cluster`.

> **NOTE** **Windows SmartScreen or your antivirus may warn about the setup program the first time.** The program is new and is not signed with a paid code-signing certificate, and some security programs are cautious about files they have not seen before. Check that the file you downloaded is the one published on the Releases page by comparing its SHA-256 checksum (the release page lists it; in PowerShell type `Get-FileHash .\Ets2ClusterSetup-1.5.0.exe`). If your security program still flags it, you can report it to the vendor as a false positive. Please do not switch your antivirus off.

3. First start

1. Start the **TruckSim GPS Telemetry Server** and then Euro Truck Simulator 2 (the order does not matter).
2. Start **ETS2-ATS Instrument Cluster**. A window with the cluster opens in the middle of your main screen.
3. Load a profile and get into the truck. The needle, gauges and numbers come alive as soon as the game is running a drive.

Until the game is connected the cluster shows its layout dimmed with a message in the middle:

Message	Meaning
Waiting for the game	The telemetry server answers, but the game is not running or is not in a drive yet.
Telemetry server not reachable	The cluster cannot reach the telemetry server: it is not running, or it is on another address (see section 7).

4. Reading the cluster

The cluster is drawn on a 16:9 canvas and scales to any window size, so it looks the same large or small. The numbers in the picture match the list below.



The cluster with demonstration values. 1 lamps, 2 speed, 3 dial, 4 gear, 5 rev strip, 6 diesel, 7 AdBlue, 8 left panel, 9 right panel, 10 bottom row, 11 temperatures.

	What it shows
1	**Lamps**: the left and right **turn signals** (green when on) and the **low beam** (green) and **high beam** (blue) symbols. Dim symbols are off.

	What it shows
2	**Speed** in big digits, with the unit (km/h or mph). The digits turn **amber** when you are more than 3 km/h (2 mph) over the speed limit.
3	**Dial** with a red needle that glides between updates. The tick marks light up as far as your speed. A **green mark** shows the cruise-control setting and an **amber triangle** the current speed limit (red while you are over it).
4	**Gear**: for an automatic gearbox *A 12* means automatic in 12th; *N* is neutral and *R* reverse. A manual gearbox shows just the number.
5	**Rev strip**: engine speed in hundreds of rpm with a red marker (amber near the limiter).
6	**Diesel gauge** (green) with the level in per cent, *1/2* in the middle and *E* at the bottom. It turns red when the game raises its low-fuel warning.
7	**AdBlue gauge** (blue), the same way.
8	**Left panel**: *Average consumption* (the game's own figure), *This trip* (measured by the cluster since the last trip reset; shows -- until you have driven a short way) and *Range* (amber when fuel is low).
9	**Right panel**: the *Cruise control* setting (green when cruise control is on, -- when off), the *Arrival (ETA)* on the game clock for the route the game's navigation is following, and the *Speed limit*.
10	**Bottom row**: *Game time*, *Rest in* (time until the game says you need a rest; amber under 30 minutes), the **Local time** of your PC in large digits in the middle, *Trip* distance and *Odometer*.
11	**Coolant** and **oil** temperature in the top corners (red when the game raises a warning). The outside temperature is not available from the game.

5. Warnings and states

The cluster changes its look to tell you what is going on without reading any numbers.



Over the speed limit: 98 km/h in a 90 zone. The speed digits and the glow beside the dial turn amber and the limit marker turns red.



Several status tags at once: engine brake, retarder step 2, wipers, beacon, and the air, fuel and AdBlue warnings. Low diesel and AdBlue show red.

Tag	Colour	Meaning
P BRAKE	red	The parking brake is on.
ENGINE BRAKE	amber	The engine brake is on.
RETARDER n	amber	The retarder is working at step n.
WIPERS	grey	The wipers are on.
BEACON	amber	The beacon (roof light) is on.
AIR	red	Low air pressure (warning or emergency).
OIL	red	Low oil pressure.
COOLANT	red	Coolant temperature warning.
BATTERY	red	Battery voltage warning.
FUEL	amber	Low fuel (15 % or less of the tank, worked out from the fuel level).
ADBLUE (DEF in ATS)	amber	Low AdBlue / DEF (15 % or less of the tank).

6. Moving, sizing and the menu

Action	How
Move the cluster	Drag it anywhere with the mouse.
Resize	Drag the grip in the bottom right corner, or turn the mouse wheel over the cluster. The 16:9 shape is kept.
Opacity (see-through)	Hold **Ctrl** and turn the mouse wheel, or choose it in the menu.
Full screen	Double-click the cluster, or press **F11**. Double-click or **F11** or **Esc** to leave full screen.
Menu	Right-click the cluster.

The right-click menu

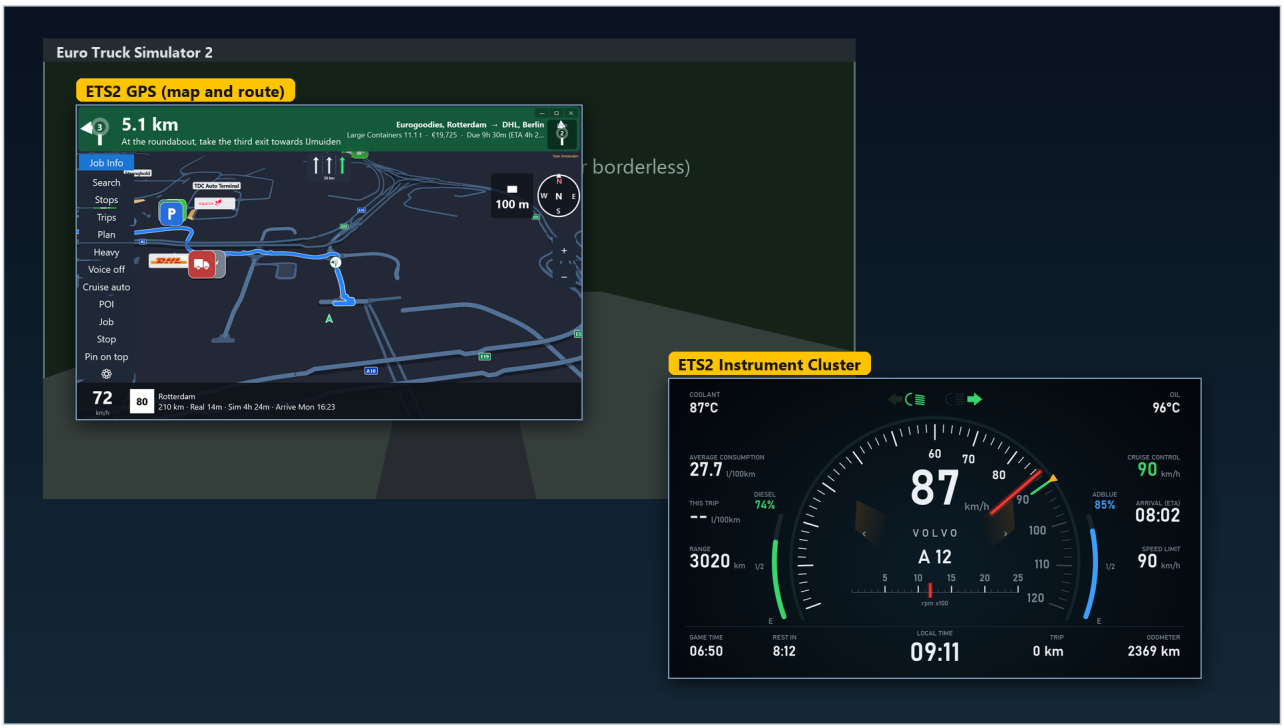
Item	What it does
Always on top	Keeps the cluster above other windows (on by default).
Opacity	100 % down to 40 %. A see-through cluster lets the game show through.
Size	Small (640 x 360), Medium (960 x 540), Large (1280 x 720), Extra large (1600 x 900) or Full screen.
Units	**Same as the game** (km/h in ETS2, mph in ATS), **Metric** (km/h, km, l/100km, °C) or **Imperial** (mph, miles, mpg, °F). This changes the speed, the dial scale, the cruise and limit values, the range, the trip, the odometer, the consumption and the temperatures.
Show the truck make	Shows or hides the make (VOLVO, SCANIA ...) under the speed.
Reset trip	Starts the trip distance and the trip consumption again from zero.
Telemetry server address...	The address of the TruckSim GPS Telemetry Server (usually 127.0.0.1:31377).
Reset window position and size	Puts the cluster back in the middle of the main screen at the medium size.
Check for updates automatically	Looks for a newer release on GitHub a few seconds after the cluster starts and then about once a day (on by default). See section 8.
Check for updates now...	Looks right away and tells you whether a newer version exists.
Minimize	Minimizes the cluster to the taskbar (click it there to bring it back). The GPS app's voice control can also minimize and restore it.
Close	Closes the cluster.

The position, size, opacity and options are remembered in `%APPDATA%\ETS2-Cluster\cluster.json` and used next time. If a monitor has been unplugged, the cluster starts on your main screen instead of off-screen.

> **TIP** The trip counter starts the first time the cluster sees your truck. Refuelling is ignored when working out the trip consumption.

7. Using it with the GPS and the game

Because the cluster is its own window, you decide where it goes. A common arrangement is the GPS and the cluster each on its own part of a second monitor, or the cluster along the bottom of the game window.



One possible arrangement: the game, the ETS2 GPS window and the cluster, each in its own place.

- Run the game **windowed or borderless** if the cluster should appear above it on the same monitor. A game in exclusive full screen hides every other window.
- Leave **Always on top** on so that the game does not cover the cluster. Use **Opacity** if you want to see the game through it.
- The cluster and the ETS2-ATS GPS app do not need each other; they both read the same telemetry server.
- To start the cluster with Windows, tick *Start the instrument cluster when I log in* in the setup program.

Telemetry server address

The cluster uses `127.0.0.1:31377`, which is right when the telemetry server runs on the same PC as the cluster. To show the cluster on **another PC**, right-click the cluster, choose *Telemetry server address...* and enter that PC's address and port 31377 (for example 192.168.1.20:31377). That PC's firewall must allow the connection.

8. Updates

The cluster can tell you when a newer version has been published. A few seconds after it starts, and then about once a day while it stays open, it asks GitHub (the public releases page of this project) which version is the newest and compares it with the one you have.



When a newer version exists, a small notice appears at the bottom of the cluster.

- The notice reads **Update x.y.z available - right-click for details**. Right-click the cluster and choose **Version x.y.z is available: open the download page** to open the release page in your browser, where you download the new setup program and run it over the old version. Your settings are kept.
- Choose **Skip this version** to hide the notice for that version. A later version shows it again.
- **Check for updates now...** in the menu looks immediately and tells you whether you have the latest version.
- Untick **Check for updates automatically** to stop all automatic checks.

> **TIP** **Nothing is downloaded or installed for you.** The check is one small request to api.github.com, which sees your IP address like any web page does; the cluster sends no information about you or your PC, and it only ever opens a page on github.com. If you are offline the check simply does nothing.

9. The Android version

The Android app shows the same cluster full screen on a tablet or phone, fed over your home network by the TruckSim GPS Telemetry Server running on the game PC. A spare tablet in landscape, standing beside the monitor, makes a convincing dashboard.



The Android cluster in demonstration mode, as shown on a tablet.

9.1 What you need

- An Android 7.0 or newer tablet or phone, on the **same Wi-Fi network** as the game PC.
- The **TruckSim GPS Telemetry Server** running on the game PC, and that PC's address on your network (for example 192.168.1.20). The PC's firewall must allow connections to port **31377**.

9.2 Installing

1. On the tablet open the **Releases** page of the ETS2-ATS-Instrument-Cluster repository on GitHub and download **Ets2Cluster-1.5.0.apk**.
2. Open the downloaded file. Android asks you to allow installing apps from this source (your browser or file manager); allow it for this one install.
3. Open **ETS2 Cluster**. It starts full screen in landscape and keeps the screen on while it runs.

> **NOTE** The app is not published in Google Play and is signed with a development key, so Android and Play Protect may ask for confirmation. Compare the file's SHA-256 checksum with the one on the Releases page if you want to be sure it is the published file.

9.3 Connecting to the game

1. **Long-press** anywhere on the cluster to open the menu, then choose **PC address**.
2. Type the game PC's address, for example `192.168.1.20` (port 31377 is assumed), and tap **Save**.
3. Start the game. The messages *Waiting for the game* and *Telemetry server not reachable* mean the same as on Windows (section 3).

To try the layout without the game, enter `demo` as the address: the cluster then shows made-up values, including a short over-the-limit state.



Demonstration mode, over the speed limit: the speed turns amber and a soft glow appears beside the dial.

9.4 The menu

Item	What it does
PC address	The address of the PC running the telemetry server, or *demo*.
Units	Tap to step through Same as the game, Metric and Imperial.
Truck name	Shows or hides the make under the speed.
Check for updates	Turns the automatic daily check on or off.
Reset trip	Starts the trip distance and trip consumption again from zero.
Check for updates now	Looks for a newer release immediately.

The local time shown in the middle of the bottom row is the **tablet's** clock. Everything else is as described in sections 4 and 5. Updates work as in section 8, except that the notice appears only for releases that include an Android file; the app opens the release page and never downloads anything itself.

> **TIP** The telemetry connection is plain HTTP on your local network, so use the app only on a network you trust.

10. Troubleshooting

Problem	What to check
Telemetry server not reachable	Start the TruckSim GPS Telemetry Server. If it runs on another PC or port, right-click the cluster > *Telemetry server address...*.
Waiting for the game	The server is running but the game is not, or is not in a drive. Start Euro Truck Simulator 2 and load a profile.
All numbers stay at zero	The game may be paused or on the menu, or the telemetry plugin is not loaded (check the telemetry server's own page).
The cluster is hidden behind the game	Run the game windowed or borderless, and make sure *Always on top* is ticked in the menu.
The cruise mark is in the wrong place	The cluster treats the cruise setting as the same unit as the speed. Tell the author if you see a mismatch.
The trip distance is odd	Right-click > *Reset trip*. After a profile or truck change the counter keeps counting from the old reading until it is reset.
The window is off-screen or the wrong size	Right-click > *Reset window position and size*. If you cannot reach the menu, close the cluster and start it again: it comes up on the main screen.
The setup program or the app is flagged by an antivirus	See the note in section 2. Compare the SHA-256 checksum with the one on the Releases page and report a false positive to the vendor.
Android: the cluster stays on *Telemetry server not reachable*	Check that the tablet and the PC are on the same Wi-Fi, that the address is right (long-press > PC address) and that the PC's firewall allows port 31377.
No update notice although a new version exists	The check runs about once a day; use *Check for updates now...* in the menu. Make sure the PC is online and that *Check for updates automatically* is ticked.

Command-line options (for testing)

Option	What it does
`--demo`	Shows made-up values instead of reading the game, to try the layout without the game. `--demo-state over` or `warn` picks other made-up states.
`--host 127.0.0.1:31377`	Uses this telemetry server address for this run.
`--config file.json`	Uses this settings file instead of the default one.
`--snapshot out.png`	Draws the cluster to a picture and quits.

Where to get it and help

- The program and this guide: github.com/johnevansau320/ETS2-ATS-Instrument-Cluster (Releases page).
- The TruckSim GPS Telemetry Server: github.com/TruckSim-GPS/trucksim-gps-server
- The ETS2-ATS GPS apps (map and route guidance): github.com/johnevansau320/ETS2GPS and github.com/johnevansau320/ETS2GPS-Win
- **Contact** (questions, problems, ideas): johnevansau320@gmail.com
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